

Distributed ledger platforms for traffic signal audit logging

A Merkle-tree append-only log — not a blockchain — is the technically correct solution for cryptographic logging of autonomous traffic signal decisions, though Hyperledger Besu offers the strongest dissertation-defensible blockchain implementation. This conclusion emerges from evaluating twelve platforms against a workload of **~37 TPS sustained** (3.2 million decisions/day across Transport for London's ~6,400 intersections) with 2 KB signed decision proofs, rare reads, and a UK municipal government operator bound by EU AI Act Article 12. The regulation is technology-neutral: it mandates automatic event recording and six-month retention but prescribes no specific tamper-evidence mechanism. (EUR-Lex) (TrueScreen) Blockchain solves a trust problem this workload does not have — the municipality is already the trusted operator — but the architectural pattern remains academically valuable. The analysis below profiles all twelve platforms, takes a position on blockchain necessity, and recommends an implementation path for a simulation-first MSc dissertation.

§1 Workload characterisation

The audit ledger records every AI signal-phase decision as a **~2 KB signed payload** containing: SHA-256 hash of input state (queue lengths, phase, waiting times), a reasoning trace, the chosen action, device ID, and model version. At TfL scale, **~500 decisions per intersection per day** across 6,400 junctions yields **~3.2 million transactions/day** — a sustained rate of **~37 TPS** with **~100 TPS bursts** during peak periods.

The critical platform properties, ranked by importance for this workload, are: **(1) deterministic finality within 60 seconds**, since decisions execute optimistically and audit records must be immutable before the next decision cycle; **(2) cost stability in GBP**, because municipal budgets cannot absorb token-price volatility; **(3) tamper-evidence via cryptographic proofs** (Merkle inclusion/consistency proofs or consensus-based finality); **(4) governance compatibility** with UK public-sector procurement and the EU AI Act's requirement for deployer-controlled log retention; and **(5) ecosystem maturity** sufficient for integration with SUMO/TraCI via Python. Throughput is effectively a non-constraint — even the slowest evaluated platform exceeds 37 TPS — making governance, cost, and regulatory fit the decisive factors.

Under EU AI Act Regulation 2024/1689, an autonomous traffic signal controller is **explicitly high-risk** (Annex III, Category 2: "management and operation of road traffic"). (EUR-Lex) Article 12 requires automatic recording of events enabling traceability, risk identification, and post-market monitoring. (EU Artificial Intelligence Act) Deployers must retain logs for **at least six months**; technical documentation for **ten years**. (Glocertinternational) (EU Artificial Intelligence Act) Crucially, the Act is technology-neutral — the terms "blockchain," "tamper-evident," and "append-only" appear nowhere in the text. A cryptographic hash chain satisfies the implied integrity

requirement. (Medium) The UK currently has no equivalent legislation, though UK-based systems sold into the EU market must comply by **2 August 2026**.

§2 Per-platform profiles

Hyperledger Fabric 2.x

Consensus and trust model. Execute-order-validate architecture with Raft ordering (crash-fault-tolerant, not Byzantine). Fully permissioned; identity via Fabric CA/MSP. Tolerates $\lfloor (n-1)/2 \rfloor$ crash failures. **Throughput.** Official Caliper benchmarks on Fabric 2.5 show ~**2,900 TPS** for blind writes with 100-byte payloads on dedicated Xeon hardware; (LF Decentralized Trust) Gorenflo et al.'s FastFabric achieved **20,000 TPS** with re-architected validation. (arXiv) The 37/100 TPS target is trivially met. **Finality.** Deterministic; configurable BatchTimeout (default 2 s). End-to-end latency 0.4–1 s under normal load. (arXiv) **Cost.** Infrastructure-only: 6 nodes (2 peers, 3 Raft orderers, 1 CA) on AWS m5.xlarge ≈ **£7.00 per million transactions** (~£670/month for 96 M tx/month). No gas fees. **Governance.** Apache 2.0 licence under LF Decentralized Trust. Channel-based policies; node admission requires consortium agreement. Multi-vendor contributor base (IBM, Oracle, Hitachi). **Tamper-evidence.** Hash-chained blocks with Merkle tree roots of transactions; full key-level history; MVCC read/write sets; HSM support for key protection. **EU AI Act posture.** Satisfies Article 12 — append-only ledger with deterministic finality, deployer-controlled retention, and export capability. Internal self-assessment conformity pathway applies. **Production deployments.** Walmart/DLT Labs freight network (70 carriers, 27 nodes, 600+ VMs); IBM Food Trust; ChitMonks regulatory audit (India). **Ecosystem.** Excellent documentation; SDKs in Go (primary), Node.js, Java; Python community-maintained. Hyperledger Caliper for benchmarking. (PubMed Central) (LF Decentralized Trust) **Risks.** Raft is CFT-only (not BFT) — acceptable for single-operator networks but cannot detect Byzantine orderer behaviour. Moderate operational complexity (CA lifecycle, channel config).

Hyperledger Besu (permissioned QBFT)

Consensus. QBFT — Byzantine fault tolerant PoA. Tolerates $<1/3$ Byzantine validators; minimum 4 nodes. (Atlassian) **Throughput.** Official wiki (May 2024): **750–1,200 TPS** on 4-node QBFT with m6i.2xlarge instances; (atlassian) research confirms scalability to 14 validators without degradation. (ResearchGate) **Finality.** Deterministic; configurable block period (default 2 s). (Hyperledger) **Cost.** 5 nodes on m6i.xlarge ≈ **£6.20–£9.15 per million tx** (~£600–£880/month). Gas price set to zero in permissioned mode. **Governance.** Apache 2.0; LF Decentralized Trust. Validator voting for admission. EEA-compliant. ~87% maintainers from Consensys. **Tamper-evidence.** Full Ethereum data model — Modified Merkle Patricia Trie for state/transactions/receipts; BFT consensus ensures irreversibility. **EU AI Act posture.** Strong — deterministic finality, full receipt logs, archive-node capability for indefinite retention. **Production.** Nigeria eNaira CBDC; LACChain (IDB-backed government network, Latin

America); mBridge (cross-border CBDC). **Ecosystem.** Full Ethereum toolchain — web3.py, Hardhat, Remix, Solidity. Excellent documentation. Runs **9.5–16% of Ethereum mainnet** nodes. **Risks.** Consensus contributor concentration; deprecated built-in permissioning (moving to plugin-based). EVM overhead is over-engineered for simple logging.

Hyperledger Iroha 2

Consensus. Sumeragi BFT ([LF Decentralized Trust](#)) (inspired by B-Chain); 3f+1 validators needed. ([Readthedocs](#)) **Throughput.** No published benchmarks for Iroha 2. Iroha 1 measured at 40–180 TPS for small networks; ([IEEE Xplore](#)) 37 TPS is plausible but unverified. **Finality.** Deterministic BFT. **Cost.** 4 validators on m5.large ≈ **£2.30 per million tx** — lowest infrastructure cost of the Hyperledger family. **Governance.** Apache 2.0 under LF; but development is **dominated by Soramitsu** (~8 Rust developers). **Tamper-evidence.** Hash-chained blocks with SCALE-encoded transactions. **EU AI Act posture.** Would satisfy Article 12 in principle. **Production.** MVP released **February 2025**; ([Hyperledger TOC](#)) deployments are exclusively CBDC PoCs (Papua New Guinea, Solomon Islands). ([LF Decentralized Trust](#)) No audit-logging deployments. **Ecosystem.** Rust (native), Python, JS SDKs; documentation improving. ([Iroha](#)) ([LF Decentralized Trust](#)) Already developing Iroha 3 (SORA Nexus), raising continuity concerns. ([GitHub](#)) **Risks.** ● **High.** MVP maturity, single-vendor dependency, no benchmarks, and potential pivot to Iroha 3. Not recommended for production or academic work requiring reproducible performance claims.

Azure Confidential Ledger (CCF)

Consensus. Modified Raft with Intel SGX TEEs; 3+ enclave nodes. CCF supports both CFT and BFT configurations, ([Wikipedia](#)) formally verified in TLA+. ([arXiv](#)) **Throughput.** CCF benchmarks: **≥65,000 req/s** (C++, 5 nodes); ([arXiv](#)) managed ACL service throughput undisclosed but trivially exceeds 100 TPS. **Finality.** Deterministic; sub-second. **Cost.** Flat rate **~£2.38/day** (~£71/month) including first 100 GB; ([Microsoft Community Hub](#)) storage overage for ~192 GB/month adds ~£15–£36. **Total** **~£86–£108/month** for 3.2 M tx/day — **per-transaction cost is effectively negligible** (~£0.000001). **Governance.** Microsoft manages infrastructure but SGX enclaves exclude Microsoft from the TCB. CCF is open-source (Apache 2.0) ([Confidential Ledger](#)) and self-hostable, providing vendor exit. Listed on **UK Government Digital Marketplace (G-Cloud)**; UK South/UK West regions available. **Tamper-evidence.** Merkle tree with periodically signed roots anchored in SGX hardware root of trust; ([Confidential Ledger](#)) offline-verifiable transaction receipts. **EU AI Act posture.** Excellent — hardware-backed integrity, deterministic finality, receipt-based verification, SOC 2 Type II + ISO 27001. ([Microsoft Community Hub](#)) **Production.** Microsoft Project Odyssey (hardware supply chain integrity); ([Microsoft Azure](#)) HB Antwerp (diamond provenance); Azure SQL Ledger digest store. **Ecosystem.** Python SDK (GA), .NET, Java, JS. No Go/Rust SDK — REST API fallback. **Risks.** Intel SGX end-of-life concerns (CCF migrating to AMD SEV-SNP); managed service TPS limits undocumented; automatic regional-pair replication (UK South ↔ UK West, acceptable). ([Microsoft Learn](#))

Amazon QLDB

⚠ **QLDB reached end of support on 31 July 2025 and is fully discontinued.** (AWS re:Post) No new ledgers can be created; existing data has been deleted. (Alvaroduran) AWS recommends Aurora PostgreSQL, which provides no cryptographic verification. **QLDB is eliminated from consideration.** Historical costs (~£93/month for the target workload) and its SHA-256 Merkle journal were competitive, but the deprecation catastrophically validates the **vendor lock-in risk** of proprietary managed ledgers. The recommended replacement loses all immutability guarantees (InfoQ) — a cautionary case study for any platform selection.

Solana

Consensus. Tower BFT + Proof of History; permissionless. **Throughput.** Vendor claims 65,000 TPS; independent measurements (ACM Digital Library) (Chainspect, April 2026) show **~978 TPS actual**, (Chainspect) with non-vote user transactions at **~1,000–1,400 TPS**. (Coinbureau) **Finality.** Probabilistic until "rooted" at **~12.8 s**; (Chainspect) effectively deterministic thereafter. **Cost.** Base fee ~£0.00031/tx; daily cost for 3.2 M tx ≈ **£992–£11,200/day** depending on priority fees. SOL has exhibited a **40× price range** over three years (£6–£241). **Governance.** Solana Foundation (Geneva); off-chain coordination; (Chainspect) municipality would have negligible influence (<0.13% of validator set). **Tamper-evidence.** Strong after finality — PoH chain + ed25519 signatures + Merkle-structured account state. No safety failures in history (all outages were liveness). **EU AI Act posture.** Technically satisfies logging requirements; however, permissionless chain governance conflicts with deployer control obligations. **Production.** Financial only — JPMorgan, BlackRock BUIDL, Visa USDC settlement. No government audit-logging deployments. **Ecosystem.** Rust (primary), TypeScript, Python (solders). 10,930 developers. (Chainspect) **Risks.** ● **Critical for this use case:** (1) 2 KB payload exceeds Solana's **1,232-byte transaction size limit** — requires splitting or hash-only storage; (2) **7+ major outages** since 2020, (helius) including 17-hour and 18.5-hour halts; (3) token cost volatility is incompatible with municipal budgeting; (4) no governance control.

Polygon PoS

Consensus. Heimdall (CometBFT/Tendermint) + Bor block production; checkpoints to Ethereum L1. **Throughput.** Vendor claims 65,000 TPS; Chainspect measures **92 TPS actual** (capacity >1,000 post-Bhilai hardfork). Vendor claims are **700–1,700× above independent measurements**. (Chainspect) **Finality.** **~5 s** (Chainspect) post-Heimdall v2 (July 2025); (Messari) additionally anchored to Ethereum every ~30 min. **Cost.** At 30 gwei gas price with 2 KB calldata: ~£0.000134/tx; daily ≈ **£429/day** (£157K/year). POL exhibited a **17× price range** and has fallen 91% from its 2024 ATH. **Governance.** Polygon Labs-driven; 105-validator cap; Nakamoto coefficient = **4** (critically low). (Chainspect) **Tamper-evidence.** CometBFT finality + dual-layer Ethereum L1 checkpoint anchoring. **EU AI Act.** Dual-layer anchoring is strong evidence, but municipality has zero governance control. **Production.** German eWpG securities, Deutsche Telekom validator, (CoinLaw) Mastercard integration. No government audit-logging. **Ecosystem.** Full EVM toolchain; 1,789 developers. (Chainspect) **Risks.** Nakamoto coefficient of 4 represents genuine centralisation; token volatility; no governance voice for municipal operator.

Avalanche L1 (custom subnet)

Consensus. Snowman (Avalanche consensus family) — probabilistic BFT that converges with overwhelming certainty. Post-Avalanche9000 (December 2024), subnets are independent "Avalanche L1s" with sovereign validator sets. (Avalanche Builder Hub) (Avax.network) **Throughput.** Custom L1 with 5-8 PoA validators: **several hundred to 1,000+ TPS** (configurable gas limits/block size). C-Chain real avg: 36.7 TPS (demand-limited). (Bitquery) (Avax.network) **Finality.** **1-2 seconds** deterministic-equivalent; sub-second claimed for custom L1s. (Avax.network) **Cost.** A permissioned L1 can set gas to zero (spam protection via permissioning) and use custom tokens. (Avax.network) **Infrastructure:** 5-8 validators $\approx$ **£6,000-£21,000/month** ($\sim$ £200-£700/day). AVAX exposure reduced to **1-10 AVAX/month per validator** for P-Chain registration ($\sim$ £35-£560/month at current prices). (Avalanche Builder Hub) **Governance. Full municipal sovereignty** — operator controls validator admission, upgrades, fees, and permissioning. No external DAO involvement. (Avalanche9000) **Tamper-evidence.** Hash-chained blocks with optional anchoring to public C-Chain via Interchain Messaging. **EU AI Act.** Excellent — sovereign governance aligns with deployer control obligations. **Production. Deloitte "Close As You Go"** — FEMA disaster recovery platform for US state/local governments using Avalanche for "tamper-evident event logging." (Commercializingblockchain) J.P. Morgan Evergreen; Mastercard. **Ecosystem.** AvalancheJS, AvalancheGo, HyperSDK; 808 developers; (Chainspect) AvaCloud managed service. **Risks.** Small validator sets halt if $> \frac{1}{3}$ offline; operational complexity; AVAX exposure is small but non-zero.

Arbitrum Nova (AnyTrust)

Consensus. Optimistic rollup (Arbitrum) with **6-member Data Availability Committee** (Arbitrum) (5-of-6 threshold; (Thestandard) members include ConsenSys, Google Cloud, QuickNode). Centralized sequencer (Arbitrum Foundation). (Thestandard) (Arbitrum) Fraud proofs via BoLD protocol. **Throughput.** Theoretical: 40,000 TPS; (Eco) actual observed: **$\sim$ 0.05 UOPS** (L2BEAT, April 2026) — near-zero current usage. Max ever: 54.89 UOPS. (L2BEAT) **Finality.** Soft (sequencer): $\sim$ 0.25 s (trust sequencer). Hard (L1): $\sim$ 14 min. Dispute window: **6 days 8 hours**. **Cost.** Per-tx: $\sim$ £0.0005-£0.005; daily $\approx$ £1,600-£16,000. All ETH-denominated. **Governance.** Arbitrum DAO (ARB token holders) controls both Arbitrum One and Nova. (Arbitrum) Municipality has **zero governance power**. **Tamper-evidence.** State roots posted to Ethereum L1; but full transaction data stored by 6-member DAC, not on-chain. (Arbitrum) **EU AI Act.** Long-term data availability depends on DAC members — not suitable for multi-year retention obligations. **Production.** Gaming (TreasureDAO). Reddit Community Points (discontinued). No government deployments. **Ecosystem.** Full EVM; Stylus (WASM). **Risks.** ● Centralized sequencer with 78-minute outage history; (Thestandard) DAC introduces trust assumptions; (Arbitrum) (Ankr) near-zero usage means untested at sustained load; ETH volatility; no governance control. Not recommended.

IOTA Rebased (formerly IOTA 2.0)

⚠ **The IOTA referenced in this evaluation has fundamentally changed.** IOTA 2.0 (leaderless

DAG/Tangle with Coordicide) was abandoned; the iota-core repository was archived ([GitHub](#)) January 2025. **IOTA Rebased** launched May 2025 as a **Move-based object ledger** ([IOTA Blog](#)) ([IOTA](#)) **forked from Sui**, using Mysticeti DPoS BFT consensus. It is **no longer a DAG, no longer feeless** — it charges gas fees (~0.005 IOTA/tx). ([IOTA](#)) Starfish consensus reached **testnet only** (v1.16.0, Q1 2026). ([IOTA Blog](#)) ([Daily News](#)) The Coordinator has been removed. ([IOTA Blog](#)) **Throughput.** Vendor claims 50,000+ TPS ([CryptoNinjas](#)) ([IOTA Blog](#)) (extrapolated from Sui); **no independent mainnet benchmark exists. Finality. ~400–500 ms** (consistent with Sui). **Cost.** Gas Station mechanism sponsors user fees: operator funds a wallet. At 3.2 M tx/day × 0.005 IOTA/tx at ~\$0.10/IOTA: **~£461K/year** — highly volatile. **Governance.** IOTA Foundation (German Stiftung); UK government engagement (Teesside digital trade testbed, 4 UK officials seconded). ([MEXC](#)) **Tamper-evidence.** BFT + checkpoint hash chain; validators sign effects certificates. **EU AI Act.** Technically satisfactory; UK government engagement is a positive signal. **Production.** UK Teesside Port/Airport digital trade (live); ([IOTA Blog](#)) Kenya trade (TWIN/TLIP, live). ([MEXC](#)) **Ecosystem.** TypeScript, Rust, Move SDKs; GraphQL API. **Risks. Four protocol rewrites** (Tangle → Chrysalis → Stardust → Rebased); mainnet only ~1 year old; 50K TPS claim unverified; token price exposure; Gas Station adds operational complexity.

peaq Network

Consensus. Substrate-based ([Bitstamp](#)) Polkadot parachain; DPoS collators + Polkadot relay chain NPoS ([Bitstamp](#)) ([Polkadot Ecosystem](#)) (GRANDPA finality). ([Polkadot Wiki](#)) **Throughput.** Vendor claims 10,000 TPS; ([Peaq +2](#)) on-chain data (Subscan) shows **~0.065 TPS average** since launch (3.8 M total extrinsics across 9.7 M blocks). ([Subscan](#)) No independent benchmarks exist. **Finality.** ~12–60 s (relay chain dependent). ([HackMD](#)) **Cost.** ~\$0.00025– ([Polkadot Ecosystem](#)) \$0.005/tx. Daily for 3.2 M tx: **~£231K–£922K/year. PEAQ tokens mandatory** — no Gas Station or sponsorship mechanism. **Governance.** EoT Labs; token governance "not immediately implemented at launch" ([Messari](#)) per Messari. **Tamper-evidence.** Relay chain finality provides strong guarantee once GRANDPA finalises. **EU AI Act.** Technically adequate but governance immaturity and token dependency are problematic. **Production.** Silencio ([Medium](#)) (noise monitoring), ELOOP ([Bitstamp](#)) (car-sharing). **Zero government deployments. Ecosystem.** 44 GitHub stars, 11 contributors. ([GitHub](#)) JS/Python SDKs. **Risks.**  **Critical:** mandatory token acquisition is a procurement deal-breaker for "no token economy"; near-zero on-chain activity; unverified TPS claims; immature governance; DePIN marketing exceeds demonstrated capability. **Not recommended for this use case.**

Trillian Tessera (CT-style Merkle log)

Architecture. Append-only Merkle tree per RFC 6962 / C2SP tlog-tiles specification. Tessera (successor to Trillian v1, which is now in maintenance mode) is a **Go library** — single binary, no microservice dependencies. ([Go Packages](#)) Entries are sequenced, integrated into the Merkle tree, and a Signed Tree Head (checkpoint) is published, optionally countersigned by independent witnesses. **Throughput.** POSIX backend: **1,587 QPS sustained**; ([GitHub](#)) MySQL: 400–500 entries/s; ([GitHub](#)) TesseraCT: up to 10,000 QPS. ([Transparency](#)) Even MySQL provides **10×**

headroom over sustained load. **Finality.** SCT (signed promise) returned synchronously; tree integration within **100 ms–2 s** (sequencer interval). **Cost.** Self-hosted open-source (Apache 2.0). Application server + MySQL + storage: **£350–£600/month** (~£4,200–£7,200/year). **No per-transaction fees, no tokens.** This is **50–100× cheaper** than the cheapest blockchain option. **Governance.** Municipality controls everything. No protocol governance, no validators, no token holders. **Tamper-evidence.** $O(\log n)$ inclusion proofs and consistency proofs — mathematically proven append-only property. However, **the operator could present split views** to different parties; mitigation via Tessera's built-in witness protocol (C2SP Witness Protocol). ([GitHub](#)) **EU AI Act.** Satisfies Article 12 — automatic recording, cryptographic integrity, deployer-controlled retention. The regulation does not require multi-party consensus. **Production.** Google Certificate Transparency (~**10.9 billion certificates in Q1 2026**, ~1,407/s average); ([Technologychecker](#)) Sigstore/Rekor (software supply chain, Linux Foundation); Go module checksum database; Google Pixel Binary Transparency. **Ecosystem.** Go library; POSIX/MySQL/GCP/AWS storage backends; built-in witnessing; well-documented. ([Transparency](#)) **Risks.** Single-operator trust model — the municipality is both operator and audited entity. External witnessing mitigates but does not eliminate this conflict. No built-in multi-party consensus.

§3 Comparison matrix

Platform	Consensus	TPS (benchmarked)	Finality	Cost/day (3.2M tx, GBP)	Gov't governance fit	Tamper- evidence	AI A 12
Fabric 2.x	Raft (CFT)	2,900	Det., ~1 s	~£22 (infra)	✓ Excellent	Hash chain + Merkle	✓ S
Besu (QBFT)	QBFT (BFT)	750–1,200	Det., ~2 s	~£20– £29 (infra)	✓ Good	MPT + BFT	✓ S
Iroha 2	Sumeragi (BFT)	Unverified	Det., est. <2 s	~£7 (infra)	⚠ Single- vendor	Hash chain + BFT	✓ Ade
Azure ACL	Raft+SGX (CFT/BFT)	≥65K (CCF)	Det., sub-sec	~£3.50 (flat)	✓ G-Cloud listed	Merkle + SGX root	✓ Exc
QLDB	Centralised	N/A	N/A	N/A	⊖ Deprecated	N/A	N/A

Platform	Consensus	TPS (benchmarked)	Finality	Cost/day (3.2M tx, GBP)	Gov't governance fit	Tamper- evidence	AI A 12
Solana	Tower BFT+PoH	~1,000–1,400 (user)	Prob., ~12.8 s	~£992– £11,200	✗ No control	PoH + ed25519	⚠ Gov gap
Polygon PoS	CometBFT+PoS	92 (obs.); >1K (cap.)	Det., ~5 s	~£429– £6,400	✗ No control; Nak. coeff. 4	BFT + L1 anchor	⚠ Gov gap
Avalanche L1	Snowman (BFT)	Custom: 100– 1,000+	Det., ~1–2 s	~£200– £700 (infra)	✓ Full sovereignty	Hash chain + opt. L1 anchor	✓ Exc
Arbitrum Nova	Optimistic+DAC	~0.05 UOPS (obs.)	Soft ~0.25 s; hard ~14 min	~£1,600– £16,000	✗ DAO- governed	L1 state roots; DAC data	⚠ I rete risk
IOTA Rebased	Mysticeti (BFT)	Vendor: 50K (unverified)	Det., ~400 ms	~£1,263 (token)	⚠ Foundation- led; UK engagement	BFT + checkpoints	✓ Ade
peaq	DPoS+Polkadot NPoS	0.065 (obs.); 10K (claimed)	~12–60 s	~£633– £2,526 (token)	✗ Token gov; no govt use	Relay chain finality	⚠ 1 dep
Tessera (CT log)	Single operator + witness	1,587 (POSIX)	Det., ~0.1–2 s	~£13– £20 (infra)	✓ Full control	Merkle proofs (RFC 6962)	✓ Suff

§4 Does this workload require a blockchain at all?

The case for blockchain

Multi-party consensus provides a cryptographic guarantee that **no single actor — including the municipality itself — can silently rewrite or censor the audit trail**. This matters because the municipality is simultaneously the log operator and the entity whose AI decisions are being audited. In a CT-style log, the operator controls writes and could theoretically omit embarrassing

entries (e.g., a signal decision that contributed to a fatal collision) or present split views to different auditors. CertLedger (Kubilay et al., 2018) demonstrated that CT logs are structurally vulnerable to split-view attacks without robust gossip. A BFT blockchain with validators from multiple parties (e.g., the municipality, TfL oversight, an independent auditor) eliminates this vector by construction — rewriting history requires corrupting $>\frac{2}{3}$ of validators.

For an MSc dissertation, implementing blockchain also makes the work more academically interesting and defensible. The contribution is not "I built a logging system" — it is "I demonstrated and evaluated a decentralised accountability architecture for autonomous infrastructure."

The case against blockchain

The trust problem blockchain solves **does not exist in this workload's operational context**. A UK municipal government operates under democratic accountability, FOI legislation, ICO oversight, and statutory audit requirements. The probability of a council IT team mounting a coordinated Merkle-tree rewrite to cover up a signal decision is vanishingly small — and if it occurred, the operational cover-up (CCTV, sensor logs, TraCI simulation records) would be far harder to execute than the cryptographic one.

Certificate Transparency has **proven the model works at extraordinary scale** — Google's CT logs recorded ~10.9 billion certificates in Q1 2026. [Technologychecker](#) Sigstore's Rekor explicitly evaluated and rejected blockchain in favour of transparency logs, citing "no token fees, no key management burden, production readiness" and arguing that blockchain decentralisation is often illusory. [sigstore](#) Tessera's built-in witness protocol (C2SP) detects split-view attacks by having independent witnesses countersign checkpoints — achieving the monitoring benefit without consensus overhead. [GitHub](#)

The cost differential is stark: Tessera costs ~**£5,000/year**; the cheapest blockchain option (Azure ACL at ~£1,100/year) is still modest, but permissioned chains like Fabric or Besu require ~**£7,000–£10,000/year** in infrastructure, and public chains with token exposure range from **£150K to £4M/year**. The EU AI Act, as established in §1, is technology-neutral and does not require multi-party consensus.

Position taken

For this specific workload — a single municipal operator logging its own AI decisions — a Merkle-tree transparency log with external witnessing is the technically correct and cost-efficient solution. The blockchain adds consensus overhead to solve a trust problem that governance already addresses. However, **a permissioned blockchain is defensible as an architectural choice** if multi-stakeholder verification is desired (e.g., TfL oversight bodies as independent validators) or if the dissertation's contribution centres on the decentralised accountability pattern rather than operational efficiency. The honest framing is: blockchain is a policy choice about who you trust, not a technical requirement of the workload.

§5 Decision matrix

Scenario (a): Maximum decentralisation and censorship resistance

Platform	Score (1-5)	Rationale
Solana	★★★★	Nakamoto coeff. 20; 760 validators; no safety failures; but 2KB limit and cost volatility
Avalanche (public C-Chain)	★★★★	661 validators; Nakamoto coeff. 26; sub-2s finality
Polygon PoS	★★	Nakamoto coeff. 4 — critically concentrated; 105 validators capped
IOTA Rebased	★★★	80 validators; Coordinator removed; but 1-year-old architecture
Besu (public mode)	★★★★★	Runs on Ethereum mainnet (800K+ validators); highest decentralisation

Winner for scenario (a): Hyperledger Besu on public Ethereum (or equivalently, any Ethereum L1 client), accepting high gas costs as the price of maximum decentralisation. If cost matters, Solana or Avalanche C-Chain.

Scenario (b): Best fit for UK municipal operator under EU AI Act

Platform	Score (1-5)	Rationale
Azure ACL	★★★★★	G-Cloud listed; UK data residency; ~£1,100/yr; SGX integrity; CCF escape hatch
Fabric 2.x	★★★★	Proven government deployments; deterministic finality; full control; operational complexity
Besu (QBFT)	★★★★	BFT; EVM ecosystem; LACChain precedent; slightly higher ops burden
Avalanche L1	★★★★	Full sovereignty; Deloitte/FEMA precedent; minimal token exposure; but higher infra cost
Tessera	★★★★★	Cheapest; simplest; satisfies Art. 12; but lacks multi-party consensus

Winners for scenario (b): Azure ACL or Tessera — ACL if the municipality wants managed service with hardware-backed integrity and UK G-Cloud procurement; Tessera if it wants maximum simplicity and cost control. Fabric and Besu are strong if blockchain is a policy requirement.

Scenario (c): Lowest overhead if "must be blockchain" is relaxed

Platform	Score (1-5)	Rationale
Tessera	★★★★★	~£5K/yr; single Go binary; RFC 6962 proofs; proven at 10.9B entries; witness support
Azure ACL	★★★★	~£1.1K/yr managed; near-zero ops; but vendor dependency

Winner for scenario (c): Trillian Tessera, unambiguously. It provides cryptographic tamper-evidence at a fraction of the cost and complexity of any blockchain option, with a decade of production validation at Google scale.

§6 Recommendation for a simulation-first dissertation

The researcher's goal is to **demonstrate the architectural pattern** — TraCI → AI decision → cryptographic ledger write → verifiable audit trail — within a SUMO simulation, not to benchmark blockchain performance at scale. Three criteria must be balanced equally:

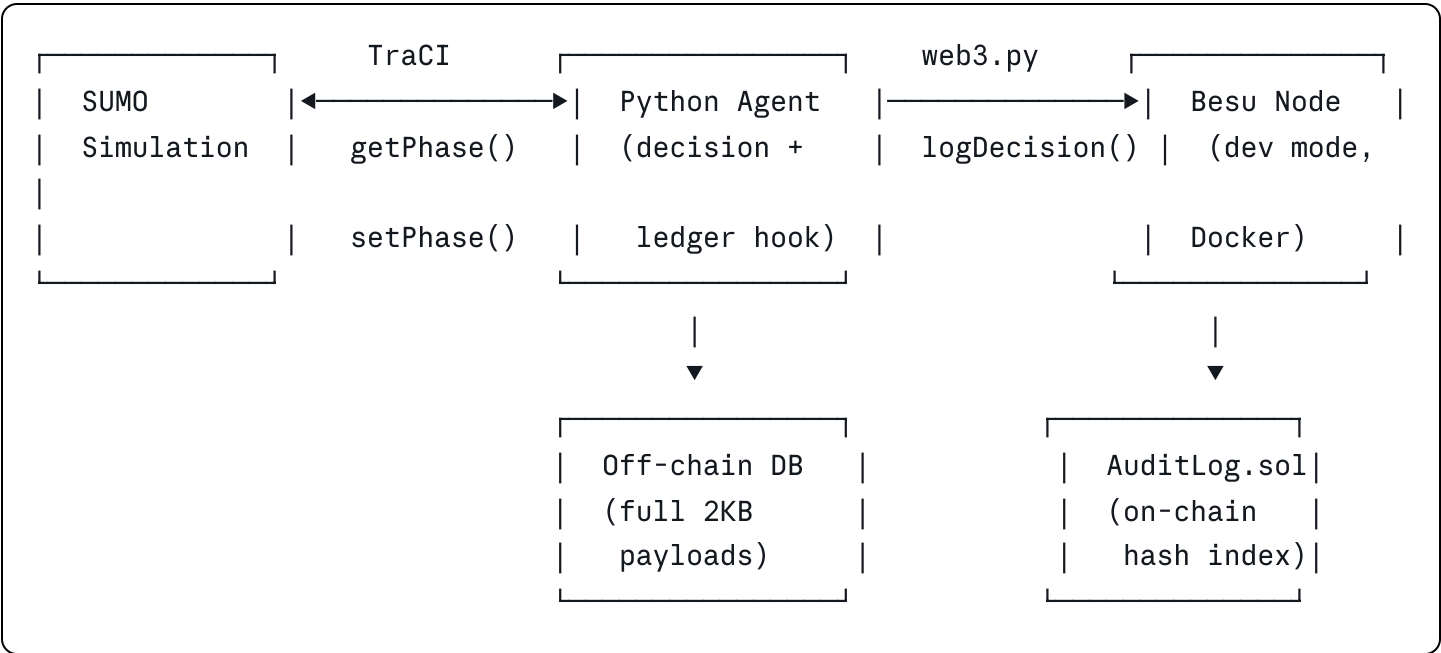
Defensibility. An MSc examiner will ask two questions: "Why blockchain?" and "Why *this* blockchain?" Hyperledger Besu in permissioned QBFT mode answers both. It has **extensive peer-reviewed literature** (Capocasale et al. 2022 benchmark Besu directly; the BLOCKBENCH framework by Dinh et al. 2017 covers Ethereum clients; LACChain provides a government deployment precedent). BFT consensus is the correct trust model for multi-stakeholder audit and stronger than Fabric's CFT-only Raft. The Ethereum ecosystem gives the examiner familiar reference points (Solidity, web3.py, block explorers). Crucially, the researcher can argue that Besu is a *credible production choice* for a permissioned government network — not a toy.

Minimal friction. Besu runs as a **single Docker container** in `--network=dev` mode, with zero configuration, instant block mining, and web3.py as the Python client — the same language as TraCI. No certificate authority, no channel configuration, no Raft cluster (all of which Fabric requires). The Solidity smart contract is trivial (~30 lines). For comparison, Tessera would be even simpler (Go HTTP client), but trades away the "blockchain" framing entirely.

Pattern clarity. The architecture is cleanly separable: TraCI loop → decision agent → ledger client → Besu node. The smart contract enforces the schema (intersection ID, decision hash, action, model version, timestamp). Transaction receipts provide Merkle proof of inclusion. The same architecture maps directly to a production Besu QBFT network by changing only the endpoint URL and adding gas estimation — demonstrating that the simulation is a faithful prototype of the real system.

Secondary recommendation. Implement a Tessera log as a **second comparator** in the dissertation. This takes ~1 day of additional work (Go HTTP endpoint + MySQL) and lets the researcher directly test the §4 question: "Does this workload need a blockchain?" Comparing Besu vs. Tessera on write latency, proof size, and operational complexity within the simulation provides original empirical evidence.

Implementation sketch: TraCI → Besu audit hook



solidity

```
// AuditLog.sol – deployed once to Besu dev network
pragma solidity ^0.8.20;

contract AuditLog {
    event DecisionLogged(
        string indexed intersectionId,
        bytes32 decisionHash,
        uint8 action,
        string modelVersion,
        uint256 timestamp
    );

    function logDecision(
        string calldata intersectionId,
        bytes32 decisionHash,
        uint8 action,
        string calldata modelVersion,
        uint256 timestamp
    ) external {
        emit DecisionLogged(intersectionId, decisionHash,
                             action, modelVersion, timestamp);
    }
}
```

python

```

# traci_audit_hook.py - main simulation loop
import traci, hashlib, json, time
from web3 import Web3

# Besu dev-mode node (single container, instant mining, zero gas)
w3 = Web3(Web3.HTTPProvider("http://localhost:8545"))
contract = w3.eth.contract(address=AUDIT_ADDR, abi=AUDIT_ABI)
ACCOUNT = w3.eth.accounts[0]

def decision_proof(tls_id, state, reasoning, action, model_ver):
    """Build 2KB signed decision payload; return hash for on-chain index."""
    payload = {
        "intersection_id": tls_id,
        "ts": time.time(),
        "input_hash": hashlib.sha256(
            json.dumps(state, sort_keys=True).encode()).hexdigest(),
        "reasoning": reasoning,
        "action": int(action),
        "model_version": model_ver,
        "device_id": f"sumo-{tls_id}"
    }
    h = hashlib.sha256(
        json.dumps(payload, sort_keys=True).encode()).digest()
    return payload, h          # payload → off-chain DB; h → on-chain

def log_to_besu(tls_id, h, action, model_ver, ts):
    tx = contract.functions.logDecision(
        tls_id, h, action, model_ver, int(ts)
    ).build_transaction({
        "from": ACCOUNT, "gas": 200_000, "gasPrice": 0
    })
    return w3.eth.send_transaction(tx)

# --- SUMO main loop ---
traci.start(["sumo", "-c", "tfl_subset.sumocfg"])
while traci.simulation.getMinExpectedNumber() > 0:
    traci.simulationStep()
    for tls_id in traci.trafficlight.getIDList():
        state = {
            "phase": traci.trafficlight.getPhase(tls_id),
            "queues": {ln: traci.lane.getLastStepHaltingNumber(ln)
                        for ln in traci.trafficlight.getControlledLanes(tls_id)},
        }

```

```

    action, reasoning = ai_agent.decide(tls_id, state)
    traci.trafficlight.setPhase(tls_id, action) # execute optimistically
    payload, h = decision_proof(
        tls_id, state, reasoning, action, "v1.0.0")
    offchain_db.store(payload) # full 2KB record
    log_to_besu(tls_id, h, action, "v1.0.0",
                payload["ts"]) # 32-byte hash on-chain
traci.close()

```

This sketch stores the **full 2 KB payload off-chain** (SQLite or flat file, sufficient for simulation) and commits only the **32-byte SHA-256 hash on-chain**, avoiding EVM calldata costs and Solana's size limits. The on-chain event index enables post-hoc verification: retrieve the off-chain payload, recompute the hash, and confirm it matches the on-chain record via `eth_getLogs`. For the Tesseract comparator, replace the `log_to_besu` call with an HTTP POST to the Tesseract personality endpoint — the rest of the architecture is identical, which is precisely the point.

§7 Explicit gaps and unverifiable claims

Missing benchmarks. No independent benchmarks exist for Hyperledger Iroha 2 throughput; the 37 TPS target is plausible but unverified. IOTA Rebased's 50,000 TPS claim is extrapolated from Sui and has no independent mainnet measurement. Peaq's 10,000 TPS claim is contradicted by on-chain data showing ~0.065 TPS average. Azure ACL's managed service TPS limits are undisclosed by Microsoft.

Unlocatable academic papers. The specific "Loghin et al. ACM DLT 2024" paper referenced in the source hierarchy could not be located in the ACM Digital Library under that exact venue/year. The closest match is Loghin et al. (arXiv:2205.06941, 2022), which benchmarks Fabric and Quorum on cloud and edge hardware. Capocasale et al. (2022, *Blockchain: Research and Applications*) was located and benchmarks Fabric, Besu, Sawtooth, and Quorum — but provides primarily qualitative comparison rather than raw TPS figures under controlled conditions.

Pricing volatility. All public-chain cost estimates are snapshots. SOL, POL/MATIC, AVAX, IOTA, and PEAQ prices are volatile; the GBP figures presented reflect April 2026 rates and could shift dramatically. Azure ACL's March 2025 price reduction to ~\$3/day may not persist.

EU AI Act harmonised standards. No CEN/CENELEC harmonised technical standard for Article 12 logging implementation exists yet (expected 2026–2027). The claim that "a hash chain satisfies Article 12" is a reasonable expert interpretation, not a tested legal position. The forthcoming ISO/IEC 24970 standard on AI system logging will provide concrete implementation guidance but is still in draft.

Tessera production readiness. Tessera was targeted for production readiness by mid-2025 and is described as beta/production-ready with alpha 3 available; Sigstore Rekor v2 GA has migrated to Tessera. However, no explicit "1.0 GA" tag was confirmed in the repository as of research date. Trillian v1 remains stable and production-proven but is in maintenance mode.

QLDB successor gap. AWS's recommended QLDB replacement (Aurora PostgreSQL) provides no cryptographic verification. No AWS service currently fills the append-only verifiable ledger niche. Organisations migrating from QLDB lose all immutability guarantees — a gap the industry has not resolved.